

Hyojoon Park

PH.D. CANDIDATE | RESEARCH SCIENTIST (3D GRAPHICS, VISION, DEEP LEARNING)

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Introduction

- My research integrates **Machine Learning** with **Computer Graphics** and **Computer Vision**, building end-to-end systems that combine physics-based simulation, geometry, and image/video processing.
- My work spans **3D reconstruction** of the face, body, hands, and hair, along with **neural rendering**, **XR (VR/AR)**, and **haptics**, with peer-reviewed publications in both Computer Graphics and Medical Imaging.
- I currently work on efficient **deep learning-based 4D medical image and video compression**, aimed at making high-dimensional data practical to store and analyze.

Education

University of Wisconsin-Madison

Wisconsin, USA

Ph.D. Candidate in Computer Sciences

Sep. 2021 - Expected Dec. 2026

- **Research area:** Synergistic integration of machine learning in physics-based simulation for computer graphics and medical imaging
- **Advisor:** Professor Eftychios Sifakis, **Co-advisor:** Professor Kevin Elceiri

Seoul National University

Seoul, Korea

M.S. in Mechanical Engineering

Mar. 2017 - Feb. 2019

- **Research area:** Rendering and transparent control of high-performance haptic system
- **Thesis:** Dental Simulator with Increased Z-width of Haptic Rendering (also presented at AsiaHaptics 2018)
- Selected for **Outstanding MS Thesis Presentation Award** [[M.S. thesis presentation](#)]
- **Advisor:** Professor Dongjun Lee

Technical University of Munich (TUM)

Munich, Germany

B.S. Exchange Student in Mechanical Engineering

Spring 2014

Korea University

Seoul, Korea

B.S. in Mechanical Engineering

Mar. 2010 - Feb. 2017

- Military service: Jun. 2011 - Mar. 2013

Work Experience

Google

Playa Vista, CA, USA

Student Researcher

Sep. 2025 - May 2026

- Developed a real-time, self-supervised neural hair simulator for digital humans, focusing on efficient physics-based animation for XR. Publication-track results in progress.

Nokia Bell Labs

New Providence, NJ, USA

Industrial Metaverse Intern

Jun. 2025 - Aug. 2025

- Developed a real-time 3D Gaussian Splatting digital twin with precise physical registration (scale and orientation) and language features for natural-language object queries, as part of "Empowering Digital Twins with Precise Physical Registration and Language-Awareness." Selected as the **1st place winner** among 100+ global interns.
- Follow-up work studies how capture geometry affects 3DGS reconstruction quality. Publication-track results in progress.

NVIDIA

Santa Clara, CA, USA

Software Intern – Physics-Based Simulations

May 2024 - Aug. 2024

- Developed a single-image 3D face reconstruction framework generalizing across ages, trained on synthetic data spanning from infants to the elderly.
- Enabled controllable re-aging and de-aging of reconstructed faces via differentiable rendering and a GAN-based objective.

University of Utah

Salt Lake City, UT, USA

Graduate Research Assistant

Sep. 2019 - Jun. 2021

- Developed and open-sourced a multi-view camera calibration framework (bundle adjustment, ML-based outlier detection) for "Capturing Detailed Deformations of Moving Human Bodies" (SIGGRAPH 2021), tracking 1,000+ uniquely labeled points to reconstruct fine skin deformations of a moving body using a low-cost patterned suit and occlusion-resilient deep networks. [[code](#)] [[paper](#)]

Korea Institute of Science and Technology (KIST)

Seoul, Korea

Research Intern

Mar. 2019 - Aug. 2019

- Developed a VR hand-rendering framework for wearable fingertip-tracking devices, reconstructing realistic full-hand poses from fingertip positions alone through motion retargeting and novel inverse-kinematics hinge constraints. Published as "Adaptive Precision-Enhancing Hand Rendering for Wearable Fingertip Tracking Devices" (IROS 2020). [[paper](#)] [[video](#)]

Publications

- **Near-realtime Facial Animation by Deep 3D Simulation Super-Resolution**

Hyojoon Park, Sangeetha Grama Srinivasan, Matthew Cong, Doyub Kim, Byungsoo Kim, Jonathan Swartz, Ken Museth, Eftychios Sifakis
ACM Transactions on Graphics (TOG), 2024 (Presented at SIGGRAPH ASIA 2024) [\[paper\]](#) [\[code\]](#) [\[featured video\]](#)

- Achieves high-resolution facial animation 115x faster (18 FPS) than traditional physics-based simulation, generalizing to unseen expressions, dynamics, and external forces.

- **Collagen Fiber Centerline Tracking in Fibrotic Tissue via Deep Neural Networks with Variational Autoencoder-based Synthetic Training Data Generation**

Hyojoon Park*, Bin Li*, Yuming Liu, Michael S. Nelson, Helen M. Wilson, Eftychios Sifakis, and Kevin W. Eliceiri (*equal contributions)
Medical Image Analysis, 2023 [\[paper\]](#)[\[code\]](#)

- Introduces DuoVAE, a dual-VAE that controls target properties during generation, producing diverse synthetic training data robust to noisy labels for collagen-fiber centerline tracking.

- **Capturing Detailed Deformations of Moving Human Bodies**

He Chen, **Hyojoon Park**, Kutay Macit, and Ladislav Kavan
SIGGRAPH, 2021 [\[paper\]](#)[\[code\]](#)

- Reconstructs fine skin deformations of a moving body (e.g., breathing and shaking) using a low-cost patterned suit, multi-camera capture, and occlusion-resilient deep networks tracking 1,000+ labeled points.

- **Adaptive Precision-Enhancing Hand Rendering for Wearable Fingertip Tracking Devices**

Hyojoon Park and Jung-Min Park
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2020 [\[paper\]](#)[\[video\]](#)

- Reconstructs realistic full-hand poses in VR from fingertip positions alone, combining motion retargeting with novel inverse-kinematics hinge constraints for real-time rendering.

- **Dental Simulator with Increased Z-width of Haptic Rendering**

Hyojoon Park, Myungsin Kim, and Dongjun Lee
AsiaHaptics, 2018 [\[paper\]](#)[\[video\]](#)

- Renders virtual teeth up to 10x stiffer (Z-width) than standard methods using commercially available haptic devices, for realistic dental feedback in VR.

- **Rigid-body Collaborative Manipulation among Remote Users with Wearable Cutaneous Haptic Interfaces**

Myungsin Kim, WonHa Lee, **Hyojoon Park**, Junghan Kwon, Yong-Lae Park, and Dongjun Lee
AsiaHaptics, 2018 [\[paper\]](#)[\[video\]](#)

- Enables remote users to jointly manipulate rigid bodies through wearable cutaneous haptic interfaces, using passivity-based control for stable and realistic interactions.

- **Stretchable Skin-Like Cooling/Heating Device for Reconstruction of Artificial Thermal Sensation in Virtual Reality**

Jinwoo Lee, Heayoun Sul, Wonha Lee, Kyung Rok Pyun, Inho Ha, Dongkwan Kim, **Hyojoon Park**, Hyeonjin Eom, Yeosang Yoon, Jinwook Jung, Dongjun Lee, and Seung Hwan Ko
Advanced Functional Materials, 2020 [\[paper\]](#)

- Reproduces both hot and cold sensations in VR from a single stretchable, skin-like device (230% stretchable), combining heating and cooling in one structure.

- **Highly Stretchable and Oxidation-resistive Cu Nanowire Heater for Replication of the Feeling of Heat in a Virtual World**

Dongkwan Kim, Junhyuk Bang, Wonha Lee, Inho Ha, Jinwoo Lee, Hyeonjin Eom, Myungsin Kim, Jungjae Park, Joonhwa Choi, Jinhyung Kwon, Seungyong Han, **Hyojoon Park**, Dongjun Lee, and Seung Hwan Ko
Journal of Materials Chemistry A, 2020 [\[paper\]](#)

- Replicates virtual heat on a wearable glove using a 12-pixel stretchable thermal array with laser-patterned copper-nanowire heaters, controlling temperature to within 5% error.

- **Wearable Cutaneous Haptic Interface with Soft Sensors and IMUs**

Yongjun Lee, Myungsin Kim, Yongseok Lee, **Hyojoon Park**, and Dongjun Lee
Korea Robotics Society Annual Conference, 2018

- **Design and Performance Evaluation of Wearable Haptic Interfaces**

WonHa Lee, Myungsin Kim, **Hyojoon Park**, and Dongjun Lee
International Conference on Control, Automation and Systems, 2018

Services

Technical Skills

Languages	Python, C++, C, CUDA
Frameworks	PyTorch, JAX, OpenCV, NumPy/SciPy, OpenGL
Core Areas	3D Vision, Neural Rendering, Physics-based Simulation, Deep Learning, Computer Graphics, 3D Reconstruction, Data Compression
Tooling	Linux, Git, Docker, SLURM, CMake, $\text{\LaTeX}$

Teaching Assistant

Computer Graphics (CS559)	<i>University of Wisconsin-Madison</i>	<i>Spring 2022</i>
Computer Graphics (CS559)	<i>University of Wisconsin-Madison</i>	<i>Fall 2021</i>
Interactive Computer Graphics (CS6610)	<i>University of Utah</i>	<i>Spring 2021</i>
Computer Graphics (CS4600)	<i>University of Utah</i>	<i>Fall 2020</i>
System Analysis in Mechanical Engineering	<i>Seoul National University</i>	<i>Spring 2018</i>